MIND MATRIX

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SESSION: 05

REPORT ON: LAB REPORT

TITLE:

CONTACT FILTER CONSOLE

STUDENT TEMPLATE

1. Introduction

This console-based Kotlin program manages and filters a list of contacts. It demonstrates the use of data classes, null handling, conditionals, and higher-order functions with lambdas. The program prints all contacts, favorite contacts, and contacts with email in a formatted structure.

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2. Kotlin Code

```
data class Contact(
    val name: String,
    val phone: String?,
    val email: String?,
    val favorite: Boolean
)

fun formatContact(c: Contact): String {
    val phoneDisplay = c.phone ?: "N/A"
    val emailDisplay = c.email ?: "N/A"
    val star = if (c.favorite) "*" else ""
    return "${c.name} | Phone: $phoneDisplay | Email: $emailDisplay | $star"
}

fun filterContacts(
    contacts: List<Contact>,
```

```
    predicate: (Contact) -> Boolean
): List<Contact> {
    return contacts.filter(predicate)
}

fun printContacts(title: String, contacts: List<Contact>) {
    println(title)
    contacts.forEachIndexed { index, contact ->
        println("${index + 1}. ${formatContact(contact)}")
    }
    println()
}

fun main() {

    val allContacts = listOf(
        Contact("Maya Patel", "+91-90000-11111", "maya@example.com",
true),
        Contact("Raj Sharma", null, "raj@sample.com", false),
        Contact("Li Wei", "+86-13800000000", null, false),
        Contact("Sara Ali", null, null, false),
        Contact("Tom O'Brien", "+1-555-1234", "tom@example.org", true)
    )
```

```
    printContacts("--- All Contacts ---", allContacts)

    printContacts(
        "--- Favorite Contacts ---",
        filterContacts(allContacts) { it.favorite }
    )

    printContacts(
        "--- Contacts with Email ---",
        filterContacts(allContacts) { it.email != null }
    )
}
```